



Orthoclase

△ **Rare, 250-carat**, yellow orthoclase gem, unusual for its size and clarity

The pink crystals of orthoclase give common granite its characteristic pink colour. It is also an important rock-forming mineral that yields gemstones. Yellow and colourless orthoclase is faceted for collectors when transparent, and it sometimes produces gems called sunstone (see p.168). A cat's-eye effect results when some yellow and white specimens are cut *en cabochon*. A variety of orthoclase exhibiting adularescence is called moonstone. This adularescence results from the interlayering of orthoclase with albite (see p.172).

Specification

Chemical name Potassium aluminosilicate | **Formula** KAlSi_3O_8
Colours Colourless, white, cream, yellow, pink, brown-red
Structure Monoclinic | **Hardness** 6–6.5 | **SG** 2.5–2.6
RI 1.51–1.53 | **Lustre** Vitreous | **Streak** White | **Locations** Myanmar, Sri Lanka, India, Brazil, Tanzania, USA, Mexico

Block-like surface lines



Orthoclase crystal | Rough | This orthoclase crystal provides a good illustration of the mineral's classic blocky shape in its natural state.

Water-wear



Gem orthoclase | Rough | In this water-worn piece of yellowish orthoclase, a high degree of transparency is readily apparent.

Textured surface



Cabochon | Colour variety | Pink orthoclase that is translucent rather than transparent is commonly cut into attractive cabochons, such as this stone.

Smoky texture



Table facet

Moonstone | Cut | This gem is a cushion, brilliant-cut moonstone (see box, left), exhibiting a characteristic silvery-white texture, emphasized by the cut's many facets.

Sacred moonstone

Legends and beliefs

Orthoclase is one of several feldspars that show a white or silvery adularescence when cut *en cabochon* and are called moonstone. Other moonstones are anorthoclase, sanidine, albite, and oligoclase (see p.164, p.172, p.168). Moonstone was sacred in India, where it was said to inflame passions, and that if lovers placed it in their mouths at full moon, their futures would be revealed. In 11th-century Europe, moonstone was believed to reconcile lovers, and in 16th-century England, a moonstone dedicated to King Edward VI was said to wax and wane with the moon.



Moonstone cabochon | This polished cabochon shows characteristic adularescence.

Background facets visible

Light refracted by cut

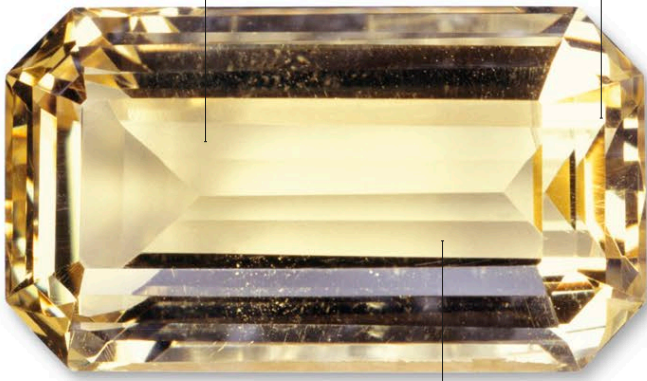


Table facet

Yellow gem | Cut | A lapidary has chosen a rectangular step cut for this yellow orthoclase stone in order to highlight its fine colour and transparency.